

A
INTERNSHIP PROJECT REPORT ON

Python

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Of
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In
COMPUTER ENGINEERING

Apollo Institute of Engineering and Technology(AIET)



Gujarat Technological University, Ahmedabad

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Apollo Institute of Engineering & Technology Ahmedabad

CERTIFICAT

This is to certify that the project report submitted along with the project entitled Python been carried out by Patel Dev Jigneshkumar under my guidance in partial fulfillment for the Diploma Engineering in Computer Engineering, Semester of Gujarat Technological University, Ahmadabad during the academic year 2023-24.

Prof. Sarthak Ruhela

Internal Guide

Prof. Kavita Rathod

Head of the Department



DECLARATION

We hereby declare that the Internship report submitted along with the **Python** submitted in partial fulfillment for the Diploma Engineering in Computer Engineering to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me At **Praxware** Technologies the supervision of **Prof. Sarthak Ruhela / Hitesh Kachhela** Nameand that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Patel Dev Jigneshkumar

Name of the Student

Sign of Student

CERTIFICATE

OF INTERNSHIP

IS PRESENTED TO

PATEL DEV JIGNESHBHAI

has Successfully completed 15 Days internship in Python at
"PraxWare Technologies".



A handwritten signature in black ink, appearing to read "Hitesh Kachela".

Hitesh Kachela
CEO



CEO

Hitesh Kachela

A handwritten signature in black ink, appearing to read "Hitesh Kachela".

**Date : 24-08-2023****TO WHOM IT MAY CONCERN**

We are pleased to inform you that **Mr. Patel Dev Jigneshkumar (2245503070665), DIPLOMA of Apollo Institute of engineering And technology, Ahmedabad** has Successfully complete their **INTERNSHIP** on **Python** at **Praxware Technologies Ahmedabad**. During this internship Period they worked on **WEB APPLICATION** Development using **Python Technology**.

Internship Title: Python**Email : devjpatel13@gmail.com****We wish them all success in their future endeavours****With Regards**

With Regards
FOR, PRAXWARE TECHNOLOGIES

PARTNER

Praxware Technologies

ACKNOWLEDGEMENT

I am very thankful for giving me the opportunity to undertake my internship training at the prestigious " **Praxware Technologies** ". It was a very good learning experience for me to have at this site.

First, I would like to thank Head of The Department **Prof. Kavita Rathod** of **Apollo Institute of Engineering and Technology** giving permission to commence this Internship.

Further, I would like to express my sincere gratitude to my Industrial Mentor **Hitesh Kachhela** for continuously guiding me at the company and dispel all my doubts with patience. Also, I am very obliged to my Internal Mentor **Prof. Sarthak Ruhela** for helping me throughout my internship and giving me a necessary suggestions and advices along with their valuable coordination. Without their continuous support it would not have been possible to complete my internship.

I own my wholehearted thanks and appreciation to the entire staff of the company for their cooperation and assistance during my internship.

I also thank my parents, friends, and all the members of the family for their precious support and encouragement which they had provided in the completion of my work.

Patel Dev Jigneshkumar

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ABSTRACT

This report is a detailed overview of my internship journey **Praxware Technologies**. During my Internship I have learned a lot about industry. I have learned to work in an industry space which not only enriched me professionally but also helped me grow personally as well. My contribution was appreciated by my supervisor and other members of the department. The career path I would be selecting for myself is quite influenced from my internship as I have had a great opportunity to practically learned about how the company works. I have tried my level best to make it meaningful by reflecting my works at the **Praxware Technologies**. Also, I have summarized my overall experience, with my learning and challenges faced as an intern.

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Chapter 1

About Company Profile

About Company

Praxware Technologies is one of the well-known It Development, Training & Placement Service provider. It offers variety of services including web and Mobile Development, Corporate Training and Placement Services with quality assurance. We provide creative solutions that not only cater to clients current but future needs as well. We are always committed to establishing long-term relationship with our esteem Clients and Business Partners. Compelled to have an in-depth understanding of our clients' needs and concerns, we provide total Business Solutions with an uncompromising level of excellence and integrity.

Praxware Technologies was established in February 2015 with the Aim to provide state-of-art products and innovative services in the field of Information Technology. At Praxware Technologies, we have the most experienced and dedicated developers and programmers who are always in accord with the fast growing technological orbit. We have designed and implemented a number of path breaking solutions that are user centric and target oriented.



Chapter 2

What I have learnt during Summer Internship

Day 1

Introduction

- History
- Features
- Setting up path
- Working with Python
- Basic Syntax
- Variable and Data Types
- Operator

Day 2

Input-Output

- Printing on screen

```
#code
```

```
print("hello World")
```

- Reading data from keyboard

```
#code
```

```
A=input("enter your data")
```

```
print(A)
```

Day 3

Conditional Statements

- **If**

```
# Example

a = 33
b = 200
if b > a:
    print("b is greater than a")
```

- **If- else**

```
#Example

a = 200
b = 33
if b > a:
    print("b is greater than a")
else:
    print("b is not greater than a")
```

- **Nested if-else (elif)**

```
#Example

a = 200
b = 33
if b > a:
    print("b is greater than a")
elif a == b:
    print("a and b are equal")
else:
    print("a is greater than b")
```

Day 4

Looping

- **For**

#Example

```
for x in range(6):
```

```
    print(x)
```

- **While**

#Example

```
i = 1
```

```
while i < 6:
```

```
    print(i)
```

```
    i += 1
```

- **Nested loops**

Loop no1:

Loop no2 :

Loop no”n”:

Day 5

Control Statements

- **Break**

#syntax,it is used to break the statement
break

- **Continue**

syntax , it is used to skip a step
continue

- **Pass**

syntax , it is use for null condition
pass

Day 6

String Manipulation

- **Accessing Strings**

```
#Example  
txt= "Hello This is Python String Example"  
  
print(txt)
```

- **Basic Operations**

```
#Example  
  
a="hello world"  
  
print(a[1])
```

- **String slices**

```
#Example  
  
b="hello word!!"  
  
print(b[2:5])
```

- **Function and Method *len()***

```
#Example  
  
b="hello word!!"  
  
print(len(b))
```

strip() Remove Only Space From start and End String

```
#Example  
  
c=" hello world "  
  
print(c)  
  
print(c.strip())
```

lower()

```
#Example  
  
d="Hello World"  
  
print(d.lower())
```

upper()

```
#Example  
  
d="Hello World"  
  
print(d.upper())
```

replace("_ ", " _ ")

#Example

d="Hello World"

print(d.replace("H", "P"))

Format Function” format() ”

#Example

age=28

str="My Name is Hitesh and i m {}"

print(str.format(age))

Day 7

Lists

- **Introduction**
- **Accessing list**

```
#Example  
  
thislist = ["apple", "banana", "cherry"]  
print(len(thislist))
```

- **Operations**

```
#Example  
thislist = ["apple", "banana", "cherry"]  
thislist.append("orange")  
print(thislist)
```

- **Working with lists**

```
# Example  
thislist = ["apple", "banana", "cherry"]  
newlist = ["mango", "pineapple", "papaya"]  
thislist.extend(newlist)  
print(thislist)
```

- **Function and Methods**

- a. `append()` - Adds a new element to the end of a list.
- b. `insert()` - Inserts a new element at a given index in a list.
- c. `extend()` - Adds a sequence of elements to the end of a list.
- d. `remove()` - Removes the first element in a list that matches a given value.
- e. `pop()` - Removes and returns the element at a given index in a list.
- f. `count()` - Returns the number of times a given element appears in a list.
- g. `index()` - Returns the index of the first element in a list that matches a given value.
- h. `sort()` - Sorts the elements of a list in ascending order.
- i. `reverse()` - Reverses the order of the elements in a list.
- j. `clear()` - Removes all elements from a list.

Day 8

Tuple

- **Introduction**

```
# Example  
thistuple=("A","b")  
print(thistuple)
```

- **Accessing tuples**

```
# Example  
my_tuple = ('p','e','r','m','i','t')  
print (my_tuple [0]) # 'p'  
print (my_tuple [5]) # 't'
```

- **Operations**

```
my_tuple = (4, 2, 3, [6, 5])  
my_tuple [3][0] = 9  
print(my_tuple)
```

- **Working**

- **Functions and Methods**

- a. `index()` - This method returns the index of the first occurrence of a specified element in a tuple.
- b. `count()` - This method counts the number of times a specified element appears in a tuple.
- c. `reversed()` - This method returns a reversed version of a tuple.
- d. `getitem()` - This method allows you to access elements of a tuple using square brackets (`[]`).
- e. `len()` - This method returns the length of a tuple.

Day 9

Dictionaries

- **Introduction**
- **Accessing values in dictionaries**
- **Working with dictionaries**
- **Properties**

Day 10

Functions

- **Defining a function**
- **Calling a function**
- **Types of functions**
- **Function Arguments**
- **Anonymous functions**
- **Global and local variables**

Day 11

Modules

- **Importing module**
- **Math module**
- **Random module**
- **Packages**
- **Composition**

Day 12

File Handling

- **Opening and closing file**
- **Reading and writing files**
- **Functions**

Day 13

Exception Handling

- **Exception**
- **Exception Handling**
- **Except clause**
- **Try ? finally clause**
- **User Defined Exceptions**

Day 14

Advance Python

- **OOPs concept**
- **Class and object**
- **Attributes**
- **Inheritance**
- **Overloading**
- **Overriding**
- **Data hiding**

Day 15

Regular expressions

- **Match function**
- **Search function**
- **Matching VS Searching**
- **Modifiers**
- **Patterns**

A Python internship is a great opportunity to learn and gain experience with the Python programming language. Python is a general-purpose, high-level programming language that is used in a wide variety of applications, including web development, data science, machine learning, and more.

During a Python internship, you will typically have the opportunity to:

- Learn about the fundamentals of Python programming, such as variables, data types, operators, control flow, and functions.
- Work on real-world Python projects, such as developing web applications, data science pipelines, or machine learning models.
- Gain experience working with Python libraries and frameworks, such as NumPy, Pandas, TensorFlow, and PyTorch.
- Collaborate with other Python developers and learn from their experience.

A Python internship is a great way to develop your Python skills and make yourself more marketable to potential employers. If you are interested in a career in software development, data science, or machine learning, a Python internship is a great place to start.

Here are some of the benefits of completing a Python internship:

- Gain hands-on experience with Python programming.
- Learn about the latest Python libraries and frameworks.
- Develop your problem-solving and analytical skills.
- Build a portfolio of Python projects.
- Network with other Python developers and professionals.
- Increase your chances of getting a job in the Python field after graduation.

If you are a student or recent graduate with an interest in Python programming, I encourage you to consider completing a Python internship. It is a great way to learn and gain experience that will help you launch your career in the tech industry

Libraries that I used in Desktop assistant

1. `pyttsx3` # For text-to-speech conversion
2. `peech_recognition as sr` # For speech recognition
3. `datetime` # For date and time
4. `wikipedia` # For accessing Wikipedia
5. `Webbrowser` # For opening web pages

I installed PIP in Desktop assistant

1. Pyaudio
2. Pyttsx3
3. speechRecognition
4. Wikipedia

code of those PIP use to install (in interpreter)

1. `pip install pyaudio`
2. `pip install pyttsx3`
3. `pip install speechRecognition`
4. `pip install wikipedia`

Code of Desktop Assistant

```
import pyttsx3
import speech_recognition as sr
import datetime
import wikipedia
import webbrowser

engine = pyttsx3.init()
recognizer = sr.Recognizer()

def speak(text):
    engine.say(text)
    engine.runAndWait()

def recognize(audio):
    try:
        text = recognizer.recognize_google(audio)
        print("You said: " + text)
        return text
    except:
        print("Sorry, I could not understand what you said.")
        return None

def wishMe():
    hour = int(datetime.datetime.now().hour)
    if hour >= 0 and hour < 12:
        speak("Good Morning!")
    elif hour >= 12 and hour < 18:
        speak("Good Afternoon!")
    else:
        speak("Good Evening!")
    speak("I am your desktop assistant. Please tell me how may I help you")

def listen():
    with sr.Microphone() as source:
```



```

        print("Listening...")
        audio = recognizer.listen(source)
        return audio
def execute(text):
    if "hello" in text.lower() or "hi" in text.lower():
        speak("Hello, I am your desktop assistant. How can I
help you?")
    elif "time" in text.lower() or "date" in text.lower():
        now = datetime.datetime.now()
        speak("The current time is " + now.strftime("%H:%M") + "
and the current date is " + now.strftime("%d %B %Y"))
    elif "wikipedia" in text.lower() or "wiki" in text.lower():
        query = text.split("wikipedia ")[-1] or text.split("wiki
")[-1]
        query= query.replace(" ", "_")
        speak("Searching Wikipedia for " + query)
        result = wikipedia.summary(query, sentences=2)
        speak("According to Wikipedia, " + result)
        speak("for more information see this page")
        webbrowser.open("https://www.wikipedia.org/wiki/"
+query)
    elif "google" in text.lower() or "search" in text.lower():
        query = text.split("google")[-1] or
text.split("search")[-1]
        speak("Opening Google for " + query)
        webbrowser.open("https://www.google.com/search?q=" +
query)
    elif 'youtube' in text.lower():
        query = text.split("youtube")[-1] or
text.split("youtube")[-1]
        speak("Opening Youtub for " + query)
        webbrowser.open("https://www.youtube.com/results?search_query="
+ query)
        speak("you can see video"+ query)
    elif 'open Whatsapp' in text.lower():

```

```

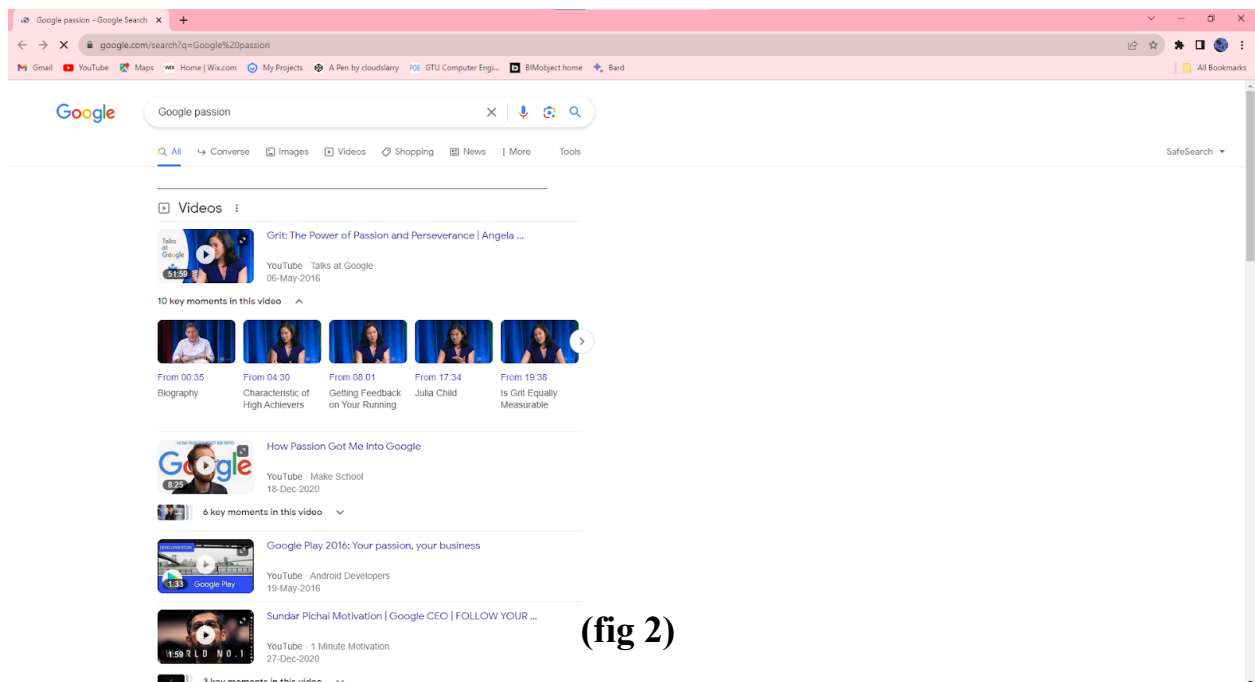
        webbrowser.open("https://web.whatsapp.com/")
    elif 'open drive' in text.lower():
webbrowser.open("https://drive.google.com/drive/my-drive")
    elif "bye" in text.lower() or "quit" in text.lower():
        speak("Goodbye, have a nice day!")
        exit()
    else:
        speak("Sorry, I did not recognize that command. Please
try again.")
if __name__ == "__main__":
    wishMe()
while True:
    audio = listen()
    text = recognize(audio)
    if text:
        execute(text)

```

Output

Listening...
You said: Google passion

(fig 1)



(fig 2)

The Python internship was a valuable and rewarding experience. I learned a great deal about Python programming, as well as about the software development process as a whole.

I gained experience with a variety of Python concepts and technologies, including:

- Data structures and algorithms
- Object-oriented programming
- Functional programming
- Web development
- Data science and machine learning

I also had the opportunity to work on real-world Python projects, which helped me to apply my skills and knowledge to solve real problems.

In addition to my technical skills, I also developed important soft skills during my internship, such as:

- Communication
- Teamwork
- Problem-solving
- Time management

Overall, the Python internship was a positive experience that helped me to grow as a programmer and a professional. I am grateful for the opportunity to have worked with such a talented and supportive team.

Here are some specific things that I learned during my internship:

- How to use Python to build web applications
- How to use Python to perform data analysis and machine learning
- How to write efficient and maintainable Python code
- How to work effectively in a team environment
- How to communicate my ideas clearly and concisely

I am confident that the skills and knowledge that I gained during my internship will be valuable to me in my future career. I am excited to continue learning and growing as a Python programmer.

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